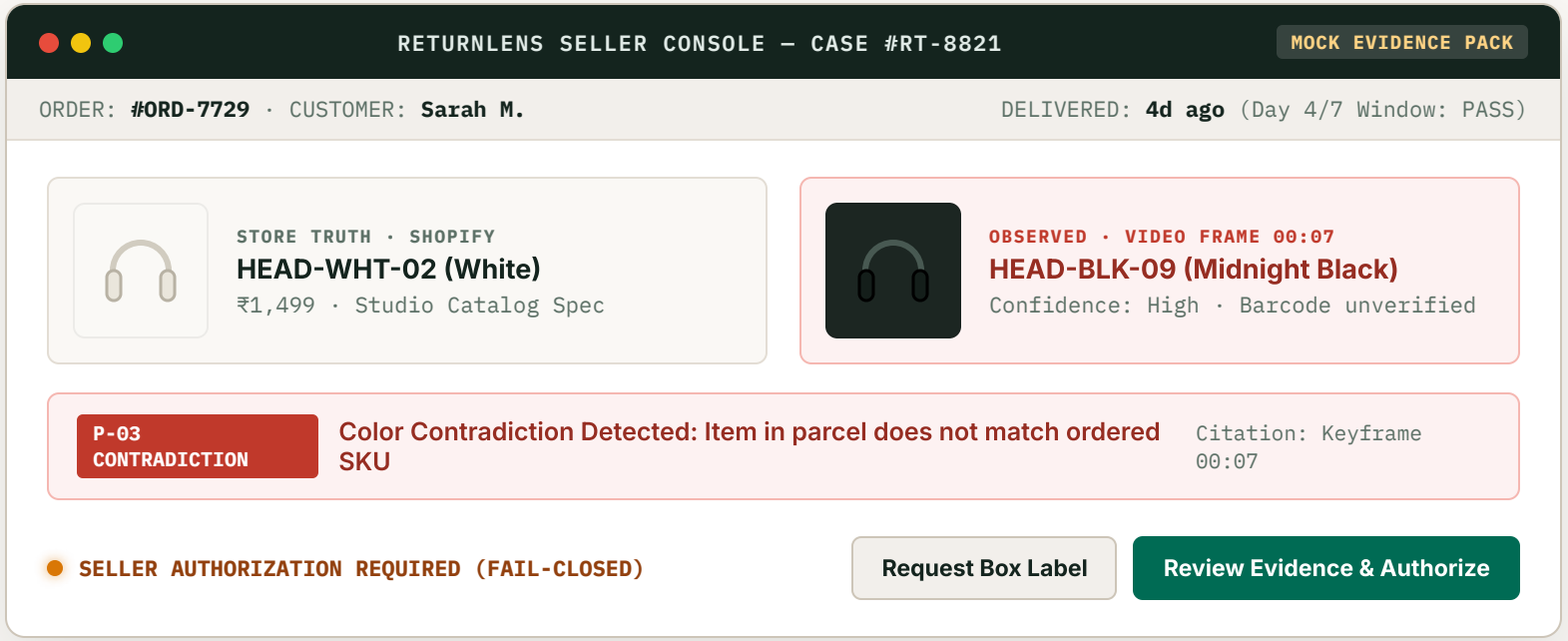
ReturnLens

Every return decision, backed by verifiable evidence.

For solo D2C merchants managing orders alone: an autonomous agent that unifies customer video, store facts, and return policy into a 1-click decision console.

> **Core Thesis:** Models observe. Code verifies. Seller authorizes.
> Scope: Return triage + order status · Mock fixtures · Never auto-refunds



THE PROBLEM SOLO MERCHANT BOTTLENECK

One return. Four disconnected systems. Zero shared record.

Solo merchants spend 15–20 minutes cross-referencing tabs from memory before moving money.

SOURCE 01 Buyer Message	Unindexed video & emotion
SOURCE 02 Store Record	Manifest ₹1,499 & SKU White
SOURCE 03 Courier API	Delivered 4d ago (in window)
SOURCE 04 Return Policy	Rules P-01 to P-08

THE CRITICAL INFORMATION GAP

“What verifiable evidence justifies moving this money?”

Without an auditable evidence pack, the seller either refunds blind (losing COGS and 2-way freight) or disputes unfairly (destroying customer trust and merchant ratings).

9% vs 15.14% **Industry Baseline:** 9% NRF 2025 criminal fraud only vs 15.14% Appriss/Deloitte 2024 fraud + policy abuse (definitions differ).

15–20 min **Asymmetric Merchant Risk:** High cognitive friction on every unboxing video triage; memory-based manual checks create financial leaks.

STAR DEMO CASE B · CORE BENCHMARK

The order says White. The unboxing video shows Black.

Policy P-03 triggers ESCALATE. No mock label created. No refund draft. No denial without human review.

ORDER MANIFEST · STORE TRUTH #ORD-7729



SKU: **HEAD-WHT-02**
Color: **Matte White**
Price: **₹1,499 paid**
Delivered: **4d ago (Shippo)**

≠

COLOR FAIL

MOCK EVIDENCE PACK · VIDEO CITATION FRAME 00:07



Observed: **HEAD-BLK-09**
Casing: **Midnight Black**
Lighting: **Optimal (142/255)**
Barcode: **Unverified**

POLICY P-03 RESULT

ESCALATE TO SELLER

3 Preserved Seller Actions: (1) Accept as warehouse mis-shipment & issue exchange · (2) Request photo of outside-box shipping label · (3) Hold return for physical parcel inspection.

Server Cap Formula:
`0 < amount ≤ min(₹4,999, rem, paid)`

SYSTEM ARCHITECTURE FAIL-CLOSED GOVERNANCE

Models observe. Code verifies. Seller authorizes.

Exactly 3 base model calls per case. Weak evidence escalates. Never releases automated refunds.

1 **Planner** Typed actions → 2 **Tools** FFmpeg ~1fps + APIs → 3 **Policy P-01-08** Deterministic rules → 4 **Critic** Review cited frames → 5 **Seller Gate** 1-Click authorization → 6 **Audit Log** Append-only JSON

CALL BUDGET

Exactly 3 Model Calls

1. Planner proposes typed tool actions.
2. Vision model inspects server-generated keyframes.
3. Critic evaluates citations against policy. Model never accesses payment APIs directly.

STATEFUL RESUMPTION

Case C: Re-Film Loop

When video is too dark (brightness 28, 2 frames), agent calls `ask_for_evidence()` with a re-film checklist. Case state saved to `WAITING_FOR_EVIDENCE` and resumes on upload (1 retry limit).

SECONDARY ROUTE

Tiny PS-04 Order Route

For routine *"Where is my order?"* inquiries, agent bypasses vision entirely: Store API → Courier API → Instant status reply. Zero token waste.

FAIL-CLOSED FINANCIAL GATE: `requires_seller_approval: true` enforced at schema layer. Server Cap: `0 < amount ≤ min(₹4,999, remaining_refundable, order_paid)`

EVALUATION SUITE 4 CONTROLLED SCENARIOS

Different evidence. Controlled, predictable action.

Every case produces an auditable mock evidence pack. Zero hallucinated policy approvals.

CORE BENCHMARK SCENARIO

ESCALATE

Case B: Color Contradiction

P-03 Failure • Order White vs Video Black

Order Record:	**HEAD-WHT-02 (White, ₹1,499)**
Observed Frame 00:07:	**HEAD-BLK-09 (Midnight Black)**
Policy Evaluation:	**P-03 Color Match: FAIL**
Automated Action:	No label • No refund draft • No denial

PRESERVED SELLER OPTIONS IN MOCK EVIDENCE PACK:

- Accept as warehouse mis-shipment & dispatch exchange
- Request customer photo of outer box shipping barcode
- Hold case for physical receipt inspection at depot

Case A: Valid Return (ORD-1042)

PREPARE SELLER REVIEW

Evidence: ORD-1042 • ₹1,499 • SKU matches catalog • Tag clearly present • Within 7-day return window.

Outcome: Mock shipping label created • Refund draft placed in pending_return_receipt (not executed).

Case C: Low-Light Video (Re-Film Loop)

ASK & RESUME

Evidence: 4-second dark unboxing video • Brightness score 28/255 • Only 2 usable keyframes extracted.

Outcome: Agent issues re-film checklist • Case stays in WAITING_FOR_EVIDENCE • Resumes on upload (1 retry limit).

Case D: Continuous Unboxing Sequence

INSPECT SEQUENCE

Evidence: Single continuous take from opening sealed parcel to displaying label in one sequence.

Outcome: Verifies frame continuity & lack of edit cuts. *Does not claim proof of identical item.*

24-HOUR PROTOTYPE PLAN TEAM ROOKIE (3 MEMBERS)

A credible, buildable prototype in 24 hours.

Strict scope discipline, mock data fixtures, and single-threaded ownership ensure on-time delivery.

00 – 06h
Foundation
Mock Store/Shippo APIs • LangGraph state schema • Policy P-01 to P-08

06 – 12h
Evidence Pipeline
FFmpeg ~1fps extractor • Vision prompt schemas • Seller Console UI mock

12 – 18h
Integration
Case C pause/resume state loop • 4 demo cases (A–D) • Critic review

18 – 24h
Validation
Fail-closed authorization tests • Boundary QA • Video demo recording

PERSON A
Backend, Policy & Audit
LangGraph state machine, policy engine P-01 to P-08, Critic validation, append-only JSON audit trail, server-side refund cap calculation.

PERSON B
Video & Seller Console
FFmpeg keyframe pipeline (~1fps sampling), vision model structured citation prompt, interactive 1-click Seller Console mockup.

PERSON C
Integration, Scenarios & QA
Test fixtures for Cases A–D, Case C stateful pause/resume lifecycle, fail-closed human gate testing, benchmark runner, video capture.

MUST DELIVER (CORE PROTOTYPE)
Mock APIs • Policy engine • Case B mismatch • Case C re-film • Seller gate • Tiny order route

CUT FIRST IF BEHIND
WhatsApp bot • Exchanges • Multi-seller auth • Audio analysis • Public cloud deployment

FINANCIAL CEILING FORMULA
$0 < amt \leq \min(₹4{,}999, rem_ref, paid)$